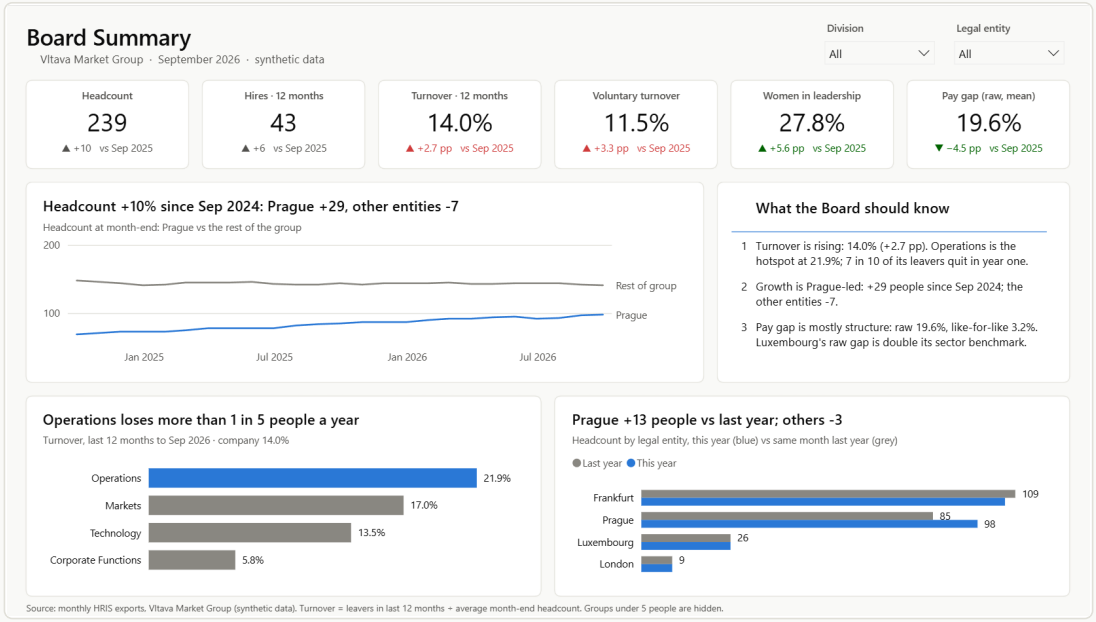
PEOPLE ANALYTICS PORTFOLIO

Kamal Salman

Four small, working demos built on one fictional company, Vltava Market Group (about 240 people in Frankfurt, Prague, Luxembourg and London), to show how I would do this job: automate the monthly people report, brief the Board, validate a reorganisation and resolve a "the numbers don't match" ticket, following written visualization standards.

Every number is tested. An independent Python calculation matches the Power BI measures on 240 out of 240 values.



Tools

Power BI (Power Query, DAX, PBIP/TMDL) · Excel (XLOOKUP, data validation) · SQL (DuckDB) · Python (pandas, matplotlib)

D1

Refresh, don't rebuild

25 monthly HRIS exports flow through Power Query into one model. A new month is one file and one click. 8 data-quality rules, each with a documented action.

"automate manual recurring reports"

D2

Board Pack in Power BI

Four pages for the Executive Board. 91 DAX measures, chart titles computed from the data, a real Eurostat benchmark, row-level security.

"reporting packages for Top Management and the Board"

D3

Reorg in a box

Excel mapping template that flags errors as HRBPs type; Python checks of spans, layers and headcount; a one-page memo.

"quarterly reorganizations across our global entities"

D4

Ticket #4711

"My list says 50, the dashboard says 47." SQL finds the three reasons; a KPI catalog stops it recurring.

"troubleshoot discrepancies" · "define new People KPIs"

Guide

Visualization standards

Message titles, one accent colour, IBCS notation, a palette tested for colour blindness, an accessibility checklist.

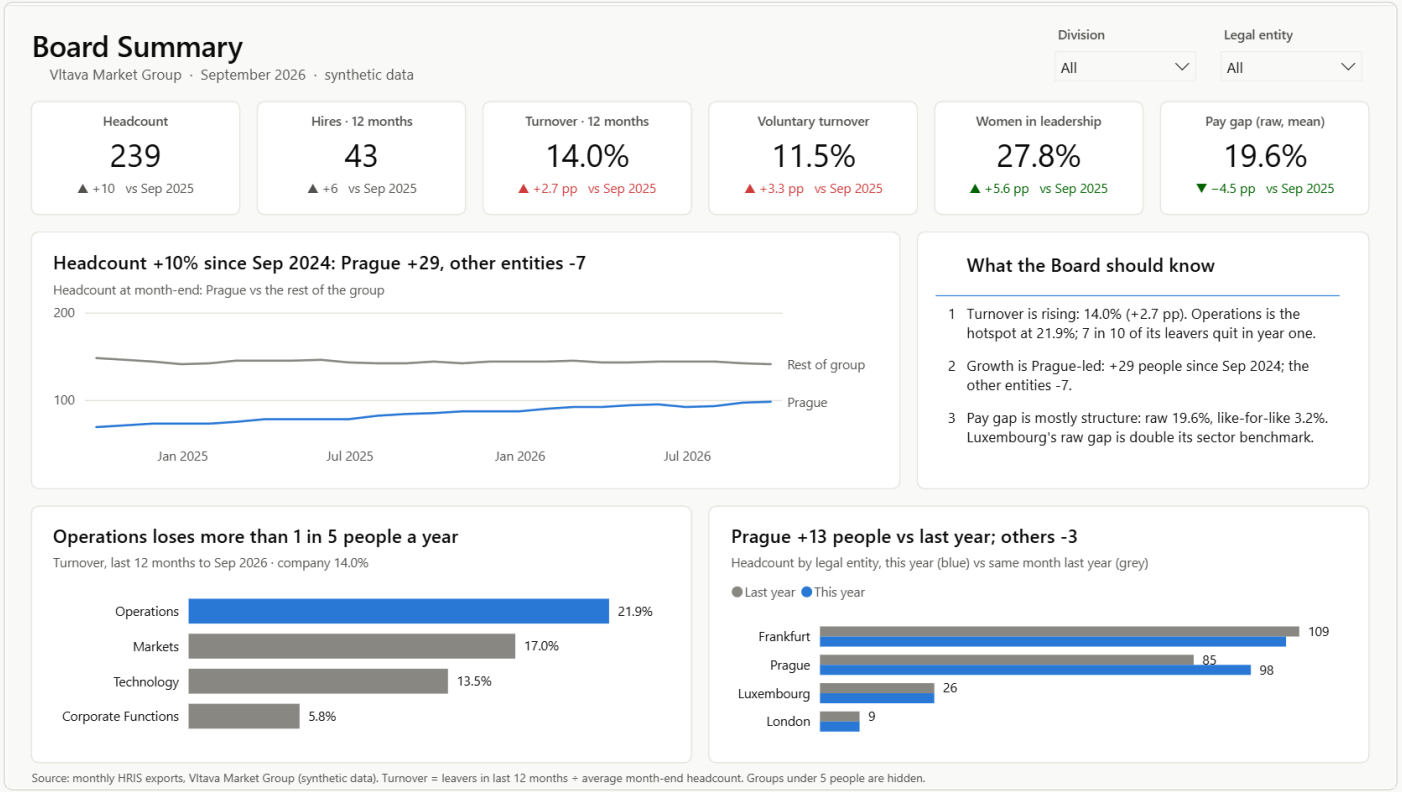
"establish best practices for data visualization"

github.com/ksalman041/people-analytics-portfolio · linkedin.com/in/kamal-salman-091411244

D2 · BOARD PACK IN POWER BI

One page tells the Board what changed

Board Summary, as rendered by Power BI Desktop after refresh (September 2026)



KPI tiles compare with the same month last year; green and red appear only when a change matters (0.5 pp or more) and always come with an arrow and a sign.

DAX

```
Turnover R12 % =
VAR _avg = [Avg Headcount R12]
RETURN IF ( [Has Full Year] && _avg >= 5,
    DIVIDE ( [Leavers R12] + 0, _avg ) )
```

DAX

```
Title Turnover by Division =
VAR _t = ADDCOLUMNS ( ALLSELECTED (
    DimDivision[Division] ), "@v", [Turnover R12 %] )
VAR _top = TOPN ( 1, _t, [@v], DESC ) ...
```

What the Board learns

- **Turnover is rising:** 14.0% over 12 months, +2.7 pp. Operations is the hotspot at 21.9%, and 7 in 10 of its leavers quit in year one, which points to onboarding, not pay.
- **Growth is Prague-led:** +29 people in two years while the other entities shrank by 7.
- **Pay gap is mostly structure:** raw 19.6%, like-for-like 3.2%; Luxembourg is double its sector benchmark.

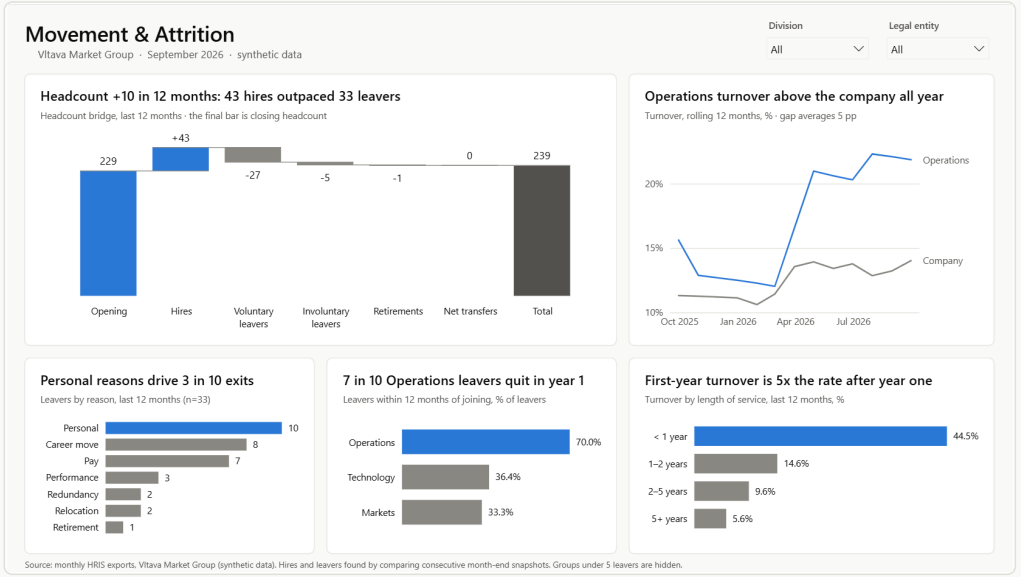
How it's built

- Star schema: month-end snapshot and movements facts; date, division, entity dimensions.
- 91 DAX measures. Rolling-12-month turnover divides by average headcount, not closing.
- Every chart title is a measure, so it can't contradict the chart.
- Row-level security role per HR Business Partner; figures for fewer than 5 people are hidden.
- Generated as a Power BI Project (PBIP): every visual and measure is text, reviewable in Git.

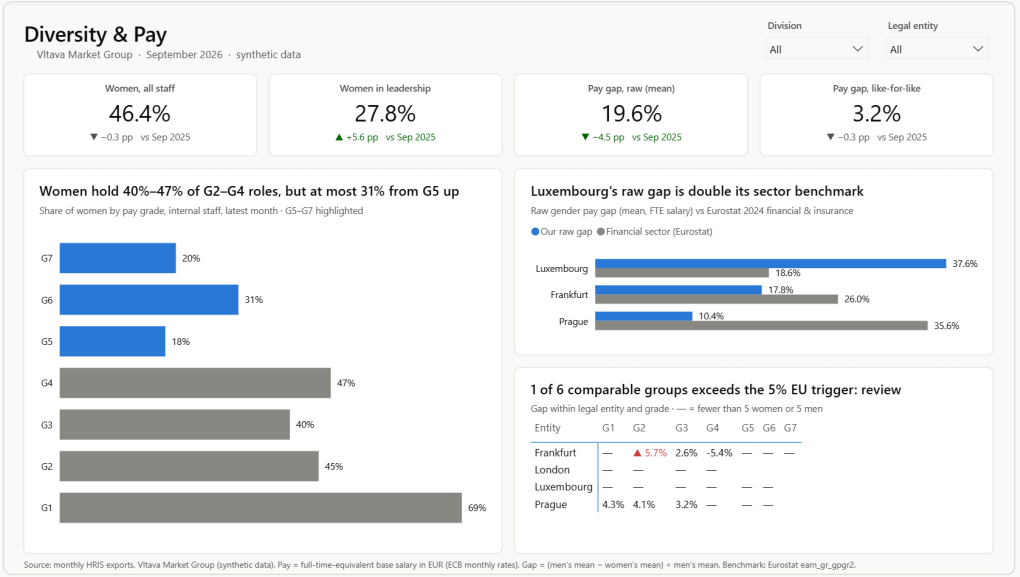
D1 · D2 · DETAIL PAGES AND AUTOMATION

Detail on demand, and a report that refreshes itself

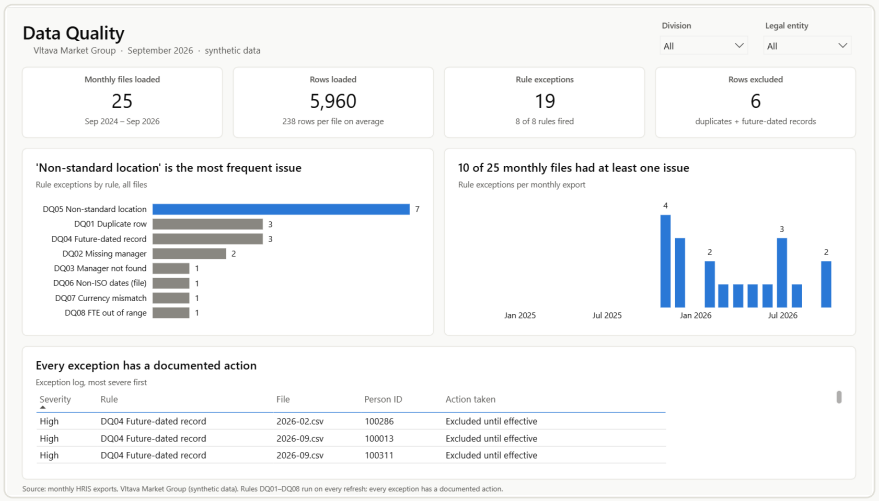
The other three pages of the Board Pack



Movement & Attrition: the headcount bridge closes exactly (229 + 43 − 33 = 239); first-year turnover is 5× the rate after year one.



Diversity & Pay: women hold 40–47% of G2–G4 roles but at most 31% from G5 up; one comparable group exceeds the 5% EU Pay Transparency trigger.



D1 · Refresh, don't rebuild

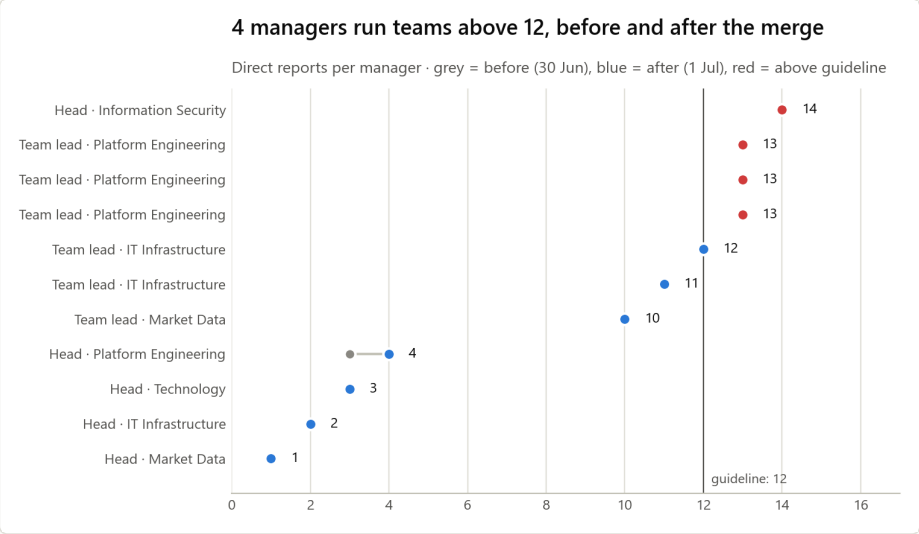
- **One folder, one click:** Power Query reads all 25 monthly exports; adding October means dropping in one file.
- **Rules, not hand fixes:** 8 data-quality checks (duplicates, future-dated records, wrong currency…), each with a severity and an action.
- **Tested:** 19 errors planted with an answer key; all 19 found, no false alarms.
- **Privacy first:** names are dropped in the first step; no KPI needs them.

Checks that run before anything goes live

Q3 2026 Technology reorganisation · Ticket #4711 from the HR Business Partner for Operations

D3 · Reorg in a box

- **First mapping blocked:** 6 issues, including a whole unit missing; 15 people would have dropped out of reporting.
- **Final mapping:** 97 positions before and after; the July HRIS matches it for 97 of 97 people.
- **For the business to decide:** the merge adds a management layer (5 → 6) for IT Infrastructure.



Structure (v2 mapping)	Before	After
Positions	97	97
Units	4	3
Average span	8.6	8.7
Spans above 12	4	4
Single-report managers	1	1
Layers (CEO to front line)	5	6

D4 · Ticket #4711: “50 on my list, 47 on the dashboard”

	People
Dashboard, Operations, 30 Sep 2026	47
+ external contractor (not headcount)	1
+ transfer into Operations effective 1 Oct (after cut-off)	1
+ new hire starting 15 Oct (future-dated record)	1
= HRBP list, pulled 2 Oct 2026	**50**

- **Method:** reproduce both numbers, anti-join the lists both ways, classify every difference, make it add up.
- **Answer:** both numbers are right; they answer different questions (employees at month-end vs everyone linked on 2 Oct).
- **Prevention:** one written definition per KPI in the catalog; future-dated records shown on the Data Quality page.

DuckDB

```
SELECT h.* FROM hrbp h
WHERE NOT EXISTS (
  SELECT 1 FROM sep_headcount s
  WHERE s.person_id = h."Person ID External"
    AND s.division = 'Operations');
```

STANDARDS · TESTING · GOVERNANCE

Written down, tested, safe with people data

From docs/viz_style_guide.md, docs/kpi_catalog.md and docs/governance.md

Visualization standards

- **Say the message:** titles are findings ("Operations loses more than 1 in 5 people a year"), not topics.
- **One accent:** blue for what matters, grey for context; status colours only on changes, with an arrow and a sign.
- **IBCS notation:** actual, previous year and plan always look the same.
- **Measured, not eyeballed:** palette checked under simulated colour blindness; text contrast at least 4.5:1.
- **Banned:** two-axis charts, gauges, 3D, pies for close values.
- **Accessible:** alt text on every chart, set tab order, a table view for every visual.

How the numbers are tested

Check	Result
DQ rules vs 19 planted errors	19 / 19
Headcount bridge closes	every month
Power BI vs pandas reference	240 / 240
Reorg v1 vs answer key	6 / 6
Reorg v2 vs July HRIS	97 / 97
Ticket reconciliation	47 + 3 = 50
Excel template (calculated by Excel)	0 errors

One definition per KPI

Turnover (rolling 12 months) = leavers in the last 12 months ÷ the average of the 12 month-end headcounts. Dividing by closing headcount would flatter a growing team and penalise a shrinking one.

Each KPI in the catalog names its population, cut-off date, DAX measure and caveats, and the same definition appears when you hover the measure in Power BI.

Governance and confidentiality

- **Synthetic people:** only Eurostat and ECB data are real, and both are public.
- **Data minimisation:** names never enter the model; pseudonymous IDs only (GDPR Art. 5).
- **Minimum group of 5:** rates, pay and diversity figures for smaller groups are hidden.
- **Least privilege:** row-level security per HR Business Partner.
- **No individual risk scores:** the EU AI Act treats AI in employment decisions as high-risk; aggregate drivers answer the question.
- **Change control:** the report is generated from code and versioned in Git.

Where to find it

docs/OVERVIEW.md: a short tour and a 5-minute demo script · powerbi/Board_Pack.pbip: the report (open, then Refresh) · docs/reorg_memo.md and docs/ticket_case.md: the two case studies · src/run_all.py: rebuilds everything from scratch